

Purva Rane

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SUMMARY

Computer Science undergraduate with strong hands-on experience in AI-driven systems, full-stack development, and real-world problem solving. Proven ability to build deployable, scalable solutions through national-level hackathons, internships, and research-focused projects. Recognized for delivering reliable outcomes under tight deadlines, collaborating effectively in teams, and applying technical skills to practical, high-impact use cases.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C

Frameworks & Libraries: React, React Native, Flutter, TensorFlow, TensorFlow Lite

Tools: GitHub, Firebase, AWS, Android Studio

Databases: MongoDB, SQLite

Core Areas: AI/ML, SDLC, Cybersecurity, Problem Solving

EXPERIENCE

Computer Hardware & Operating Systems Intern

Jun 2024 – Jul 2024

Blackbox Technologies, Pune

- Worked with computer hardware components, motherboard architecture, processors, and memory units
- Performed OS installations, system configuration, and troubleshooting
- Analyzed system failures and identified faulty or missing components
- Strengthened understanding of hardware-software interaction

PROJECTS

BreedVision – AI-Based Breed Recognition System | YOLOv12, TensorFlow Lite, Flutter

Nov 2025 – Dec 2025

- Designed and developed an AI-powered, offline-first mobile application for image-based cattle and buffalo breed recognition, optimized for low and mid-range Android devices
- Implemented a lightweight YOLOv12-based computer vision pipeline using TensorFlow Lite, enabling real-time, on-device inference without internet connectivity
- Integrated geo-tagging, image cropping, and quality checks to ensure accurate field data capture and reliable breed identification
- Adopted a human-in-the-loop workflow where the system provides top breed predictions and verified labels are stored for continuous model improvement
- Achieved over 90% accuracy during evaluations and enabled seamless integration with Bharat Pashudhan App (BPA) through structured APIs

COEP Lost & Found – Centralized Lost Item Recovery Portal | React, TypeScript, Firebase

Jan 2025 – Apr 2025

- Designed a centralized web portal as a Data Structures & Algorithms mini-project, solving real-world inefficiencies in campus lost-and-found management
- Applied efficient data organization and client-side filtering techniques to enable fast search and retrieval of lost and found items
- Developed a responsive Single Page Application using React (Vite) and TypeScript for type-safe, maintainable code
- Integrated Firebase Authentication and Realtime Database for secure login and instant data synchronization
- Implemented image-based reporting, category-wise filtering, and status tracking to improve item recovery rates
- Deployed the application on Vercel, making it live and accessible for real-world usage

WiseTrack – AI-Powered Education Platform | React, Node.js, Firebase

Aug 2024 – May 2025

- Developed an AI-driven platform to reduce student dropout rates
- Implemented early dropout prediction with 99.8% accuracy
- Presented at national hackathons including SIH 2024

Evento – Real-Time Event Management App | React Native, Firebase

Nov 2024 – Jan 2025

- Built QR-based attendance, live communication, and registration features
- Delivered 70% functionality within a 24-hour hackathon

EDUCATION

COEP Technological University, Pune <i>B.Tech in Computer Science & Engineering</i>	2025 – 2028
Dr. Babasaheb Ambedkar Technological University <i>Diploma in Computer Science & Engineering (98.80%)</i>	2022 – 2025
Maharashtra State Board <i>Secondary Education (94.40%)</i>	2022

CERTIFICATIONS

- Ethical Hacking – Internshala (98% score)
- Virtual Reality Development – Institution’s Innovation Council

ACHIEVEMENTS & ACTIVITIES

- Smart India Hackathon 2025 – Grand Finalist & Runner-Up
- 1st Rank – SHODH 2024 (Digital Forensics)
- 1st Rank – Project Presentation, TECHEXPLOER 2025
- Winner – Technical Debates (3+ years)
- Taekwondo – 1st Dan Black Belt (A+ Grade)